

PRIYADHARSINI RAMAMURTHY

☎ 331-315-1645 — ✉ pramamu@ostateemail.okstate.edu — [in LinkedIn](#) — [🌐 Website](#)

SUMMARY — Seasoned AI/ML Scientist and Engineer with 11+ years of enterprise technology experience and 6+ years of applied AI/ML research, delivering production grade intelligent solutions across healthcare, biopharma, retail, banking, and manufacturing.

SKILLSET

Programming: Python, SQL, Java, JavaScript

ML/AI: Supervised and Unsupervised Learning, Deep Learning, NLP, GNNs, XAI, Statistical Modeling, Time-Series Analysis, Feature Engineering, Model Evaluation

Generative AI: LLMs, Retrieval-Augmented Generation, Agentic AI, Prompt Engineering, Knowledge Graphs, Vector Search

Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, LangChain, LangGraph, spaCy

Data Engineering: Apache Spark, PySpark, Spark SQL, ETL Pipelines, Distributed Computing

Cloud and Deployment: AWS, GCP, Docker, Kubernetes, FastAPI, Git, CI/CD

Databases: Oracle, PostgreSQL, ChromaDB, DuckDB

PROFESSIONAL EXPERIENCE

AI/ML Engineer Intern, Oklahoma State University, Department of Veterans Affairs & Oracle

May 2026 – Present

- Delivered enterprise-scale AI, LLM, and agentic decision-support solutions across federally funded DoD, NSF, and USDA programs, advancing privacy-aware, production-ready AI in federated healthcare environments supporting 1.95M+ patients and 90K+ clinical features, with model performance reaching 0.916 AUC through rigorous evaluation, benchmarking, and privacy-aware AI governance.
- Enabled privacy-preserving local inference by translating Python ML workflows into JavaScript for browser-based deployment, retaining 100% of inference data locally and eliminating external model-serving data transfers.

AI/ML Researcher & Engineer(Research Assistant), Oklahoma State University - Center for Health Systems Innovation, Cerner

September 2025 – April 2026

- Architected LLM-powered RAG and agentic AI systems integrating vector retrieval, healthcare analytics, lightweight tool connectivity, and retrieval–reranking–critic validation, with performance targets benchmarked against literature reporting up to 8% macro-F1 and 11% recall gains. Optimized cost and latency through open-source LLM evaluation across LLaMA, Gemma, and Mistral.
- Defined evaluation frameworks, KPIs, and validation strategies spanning task success, hallucination detection, and safety for production LLM and agentic AI systems across a \$2.7M DoD-funded research program.

Data Science and Machine Learning Intern, GlaxoSmithKline - Digital Virology R&D

May 2025 - August 2025

- Developed transformer and GNN-based ML models for infectious disease incidence forecasting to support vaccine strategy. Built end-to-end pipelines in Python integrating Google Trends API and large-scale epidemiological datasets, emphasizing statistical ML, data management, and analytics. Conducted MSA and phylogenetic analysis using MAFFT, IQ-TREE, Biopython, and statistical tools for exploratory and predictive insights.

AI/ML Researcher & Engineer(Research Assistant), Oklahoma State University - Center for Health Systems Innovation, Cerner

May 2023 – April 2025

- Spearheaded scalable Spark ML/NLP pipelines on Oracle-Cerner EHR data (VA partnership). Leveraged AWS, distributed computing, and healthcare analytics to improve operational efficiency 15%, processing speed 20%, storage utilization 20%, and reduce model training time by 25%.
- Led attention based GNN models for post-traumatic epilepsy risk prediction on large-scale sparse EHR data, enhancing model interpretability, risk stratification, and clinical decision-support through graph-based AI.

AI/ML Researcher & Engineer(Research Assistant), Oklahoma State University(Computer Vision And Understanding Lab), National Science Foundation Grant

Jan 2021 – April 2023

- Designed knowledge-graph-enhanced agentic QA and decision-support systems integrating ConceptNet, dependency parsing, and POS tagging across 1M+ records. Improved answer quality by 15% through iterative knowledge integration, systematic evaluation, and BLEU-based optimization.

- Built intelligent QA agents using transformer models, retrieval pipelines, and commonsense reasoning. Drove performance gains through hypothesis-driven experimentation, ablation studies, prompt engineering, and structured error analysis.

AI/ML Researcher & Engineer(Research Assistant), Oklahoma State University(Computer Vision And Understanding Lab), United States Department of Agriculture Grant May 2020 – Dec 2020

- Developed taxonomy-aware zero-shot ML for biological sequence classification, supporting pathogen identification, genomics research, and point-of-care diagnostics in limited-label settings.

AI/ML Researcher & Engineer(Research Assistant), Oklahoma State University Aug 2019 – April 2020

- Improved fake-news detection accuracy from 87% to 93% by enhancing a propagation-path classification model using CNN and LSTM architectures.
- Developed an NLP pipeline using NER, behavioral pattern analysis, and LDA topic modeling to identify potential opioid-use risk from Reddit data.
- Built predictive NLP and statistical ML models for opioid-risk detection and diabetic patient outcomes, achieving a 20% accuracy improvement over standard benchmarks.

Senior Software Engineer(AI/ML), Global Logic, Walgreens Feb 2019 – July 2019

- Established Apache Spark parallel computing pipelines for efficient handling of large datasets, real-time analytics, and deriving actionable insights, leading to substantial improvements in memory usage efficiency and processing speed, thereby enhancing business outcomes. Utilized cloud-based resources (AWS, Google Cloud, GPU/TPU) for scalable data processing.

Senior Consultant(AI/ML), Capgemini America, McDonald's June 2017 – Feb 2019

- Served as Technical Lead within the Center of Excellence for a successful pilot product launch supporting McDonald's managed IT services.
- Led ML model deployment in microservices-based applications using Docker, Kubernetes, REST APIs, and MLOps frameworks, achieving a 25% performance improvement
- Applied CI/CD best practices and time-series analysis for business profit forecasting, with a 20% lift in digital revenue alongside the Mobile Order & Pay rollout.

Information Systems Manager Intern(Data Science), County of Santa Clara June 2016 – April 2017

- Efficiently managed five concurrent projects with distributed tasks, meeting all deadlines as a solo contributor, significantly reducing manpower costs for the county. Managed, Engineered RFPs, devised WBS for effort estimation, and oversaw status report tracking. Deployed and executed predictive modeling using Ensemble techniques and statistical ML baselines and enhanced overall prediction accuracy by 40%.

Senior Software Engineer(ML), Hyundai(Doosan) Infra Core-Bobcat March 2013 – July 2015

- As Solution Architect, led a 25-member team to launch the North America Parts Dealer Portal, applied data analysis and statistical modeling to optimize ML algorithms for derivatives pricing, contributing to a 15–20% lift in digital parts revenue across 200+ global dealers and cutting support/returns costs by over \$2M. (Excellence Award recipient)

Data Analyst, Infosys Technology Solutions December 2011 – Jan 2013

- Leveraged solid business and data analysis expertise to define and develop detailed functional requirements, process diagrams, reports. Redesigned website with angular JS, HTML Bootstrap CSS. Partnered with product owners, risk, and compliance stakeholders to align features with internal controls, audit trails, and data handling guidelines appropriate for a regulated financial services environment.

Assistant Systems Engineer, Tata Consultancy Services March 2007 – September 2011

- Achieved the highest number of defect fixes among 70 ASE trainees. As a Full Stack developer, Engineered, deployed, and validated advanced front-end (AngularJS, CSS, Bootstrap, HTML) and back-end architectures (Java/J2EE frameworks: Struts 1 & 2, Spring, Hibernate, Oracle Top Link, EJB) with remarkable efficiency, guaranteeing successful product launch.

Selected Research Publications & Scholarly Impact

Peer-Reviewed Publications

1. **Ramamurthy, P.** & Aakur, S. N. (2022). *ISD-QA: Iterative Distillation of Commonsense Knowledge from General Language Models for Unsupervised Question Answering*. Proceedings of the **26th International Conference on Pattern Recognition (ICPR)**. *CORE/QUALIS A-ranked conference, ranked among the Top 16 Computer Vision & Pattern Recognition venues by Google Scholar*. [Link](#)

2. Pallagani, V., **Ramamurthy, P.**, Khandelwal, V., Venkataramanan, R., Lakkaraju, K., Aakur, S. N., & Srivastava, B. (2022). *A Rich Recipe Representation as Plan to Support Expressive Multi-Modal Queries on Recipe Content and Preparation Process*. Proceedings of the **Knowledge Engineering for Planning and Scheduling (KEPS) Workshop**, International Conference on Automated Planning and Scheduling (ICAPS).

3. Khandelwal, V., Lakkaraju, K., Pallagani, V., **Ramamurthy, P.**, et al. (2022). *A Multi-Modal Decision Support System with Allergy-Aware Recipe Understanding Powered by a Plan Representation*. **AAAI Affiliated Demonstration/Workshop Paper**.

4. Aakur, S. N., Ramnath, V., **Ramamurthy, P.**, et al. (2024). *TEPI: Taxonomy-Aware Embedding and Pseudo-Imaging for Scarcely-Labeled Zero-Shot Genome Classification*. **IEEE Journal of Biomedical and Health Informatics (JBHI)**. *One of the Top 3 journals in Health Information Management, Impact Factor: 7.7*. [Link](#)

5. **Ramamurthy, P.**, et al. (2025). *Graph Attention Network with Comorbidity Connectivity Embedding for Post-Traumatic Epilepsy Risk Prediction Using Sparse Time-Series Electronic Health Records*. **International Journal of Medical Informatics (IJMI)**. *Abstract accepted for poster presentation at the American Epilepsy Society (AES) Annual Meeting, December 2025*. [Link](#)

Key Achievements & Honors

- **Outstanding Computer Science Researcher Award (2024)**, Oklahoma State University.
- **Research Fellowships:** Computer Science Fellowship (2019), Fisher Fellowship (2024), and DoD–Oracle-Cerner Research Fellowship through the Center for Health Systems Innovation (CHSI), Oklahoma State University.
- **AES Department of Defense Travel Award (2025)** for research contributions in healthcare AI and neuroinformatics.
- **Professional Development Allowance Recipient (2022)**, Oklahoma State University(Three Minute Thesis Competition).
- **STAR PERFORMER Award (2013)**, Doosan Infracore (HD Hyundai Infracore).
- **Excellent Contribution Award**, North America Parts (NA Parts) Implementation, Doosan Infracore (HD Hyundai Infracore).

EDUCATION

Oklahoma State University	3.9/4 GPA
Research Major: Natural Language Processing, Machine/Deep Learning	Ph.D. Candidate, Computer Science(Present)
Specialization: AI and Healthcare Analytics	M.S. in Computer Science(2022)
International Technological University	M.S. in Engineering Management(2017)
Anna University	B.S. in Computer Science(2006)

Professional Service

Conference Reviewer & Session Chair

- Reviewed peer-reviewed research submissions in NLP, Machine Learning, Deep Learning, and Medical Image Retrieval (MIR) for **ICPR 2022, ICMLA 2022, ICMLA 2021, ISES 2021, IEEE ASIAN 2026, JMIR 2026, ACM 2026** and **TMIR 2026**.
- **Session Chair: IEEE AIDML 2026, IEEE INCECT 2026:** Evaluated research presentations, moderated technical discussions, and provided constructive scholarly feedback as Session Chair.

Leadership & Volunteering

- Women in Data Science (WiDS) Ambassador (2024), assisted the Oklahoma State University Computer Science Grandparent University Program.
- Volunteer, European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD 2020).

Professional Memberships & Recognition

- **Distinguished Fellow**, Soft Computing Research Society (SCRS), 2026, **Graduate Student Member**, IEEE, 2026–2027.